Grouping of women with respect to their age is essential, since capacity of women to give birth to child varies over the age of women. Thus, *ASFR* enables to compare the fertility rates of two or more different regions with respect to specific age groups. Moreover, *ASFR* can be considered as a probability value.

Example 8.13

The female population, at reproductive age, of a country is grouped into six age groups. The number of women in each group and the number of live births given by them are given the following table.

Age Group (in years)	No. of Women	No. of Live Births
15-20	1,16,610	10,668
20-25	1,13,810	17,183
25-30	1,03,130	12,722
30-35	93,500	7,283
35-40	74,120	3,656
40-45	62,900	1,340

Calculate the general fertility rate and the age specific fertility rates of the country.

Solution:

Total number of women in the country at reproductive age during the period of 't' is

$$\sum_{i=15}^{44} P_t^i = 5,64,070$$

and the total number of live births of the country during the same period is

$$B_t = 52,852$$

Therefore, the general fertility rate of the country is

$$GFR = \frac{52852}{564070} \times 1000$$

$$= 93.69 \text{ } per\ thousand.$$

The age specific fertility rates of the country can be calculated for each age group using the formula

$$ASFR(x, x+n) = \frac{B_t(x, x+n)}{P_t(x, x+n)} \times 1000$$

where

$B_t(x, x+n)$ denotes the number of live births to the women in the reproductive age group $(x,x+n)$ during the period t in the given region, and

$P_t(x, x+n)$ denotes the number of women in the reproductive age group $(x,x+n)$ during the period t in the given region.

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